# 7.6 Conditional random fields (CRFs)

- a trade-off between localization accuracy and classification performance seems to be inherent in CNNs: deeper models with multiple max-pooling layers have proven most successful in classification tasks, however the increased invariance and the large receptive fields of top-level nodes can only yield smooth responses
- as illustrated in Figure 12, CNN score maps can predict the presence and rough position of objects, but cannot really delineate their borders
- previous work has pursued two directions to address this localization challenge
- the first approach is to harness information from multiple layers in the convolutional network in order to better estimate the object boundaries
- the second is to employ a super-pixel representation, essentially delegating the localization task to a low-level segmentation method
- an alternative direction is based on coupling the recognition capacity of CNNs and the fine-grained localization accuracy of fully connected CRFs, which is remarkably successful in addressing the localization challenge, producing accurate semantic segmentation results and recovering object boundaries at a level of detail that is well beyond the reach of existing methods

# 7.6 Conditional random fields (CRFs)

- traditionally, conditional random fields (CRFs) have been employed to smooth noisy segmentation maps
- typically these models couple neighboring nodes, favoring same-label assignments to spatially proximal pixels
- qualitatively, the primary function of these short-range CRFs is to clean up the spurious predictions of weak classifiers built on top of local hand-engineered features
- compared to these weaker classifiers, modern CNN architectures produce score maps and semantic label predictions which are qualitatively different
- as illustrated in Figure 12, the score maps are typically quite smooth and produce homogeneous classification results
- in this regime, using short-range CRFs can be detrimental, as our goal should be to recover detailed local structure rather than further smooth it
- using contrast-sensitive potentials in conjunction to local-range CRFs can potentially improve localization, but still miss thin structures and typically requires solving an expensive discrete optimization problem

# 7.6 Conditional random fields (CRFs)

- to overcome these limitations of short-range CRFs, the fully connected CRF model is used
- the model employs the energy function

$$E(\boldsymbol{x}) = \sum_i \theta_i(x_i) + \sum_{ij} \theta_{ij}(x_i, x_j),$$

  where $\boldsymbol{x}$ is the label assignment for pixels
- we use as unary potential $\theta_i(x_i) = -\log P(x_i)$, where $P(x_i)$ is the label assignment probability at pixel $i$ as computed by a CNN
- the pairwise potential has a form that allows for efficient inference while using a fully connected graph, i.e., when connecting all pairs of image pixels, $i, j$
- in particular, we use the following expression:

$$\theta_{ij}(x_i, x_j) = \mu(x_i, x_j) \left[ w_1 \exp\left( -\frac{||p_i - p_j||^2}{2\sigma_\alpha^2} - \frac{||I_i - I_j||^2}{2\sigma_\beta^2} \right) + w_2 \exp\left( -\frac{||p_i - p_j||^2}{2\sigma_\gamma^2} \right) \right],$$

  where $\mu(x_i, x_j) = 1$ if $x_i \neq x_j$, and zero otherwise, which means that only nodes with distinct labels are penalized

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- the remaining expression uses two Gaussian kernels in different feature spaces; the first, "bilateral" kernel depends on both pixel positions (denoted as $p_i$) and RGB color (denoted as $I_i$), and the second kernel only depends on pixel positions
- the hyperparameters $\sigma_\alpha$, $\sigma_\beta$, and $\sigma_\gamma$ control the scale of Gaussian kernels
- the first kernel forces pixels with similar color and position to have similar labels, while the second kernel only considers spatial proximity when enforcing smoothness
- crucially, this model is amenable to efficient approximate probabilistic inference
- the message passing updates under a fully decomposable mean field approximation $b(\boldsymbol{x}) = \prod_i b_i(x_i)$ can be expressed as Gaussian convolutions in bilateral space
- high-dimensional filtering algorithms significantly speed-up this computation resulting in an algorithm that is very fast in practice

# 7.6 Conditional random fields (CRFs)

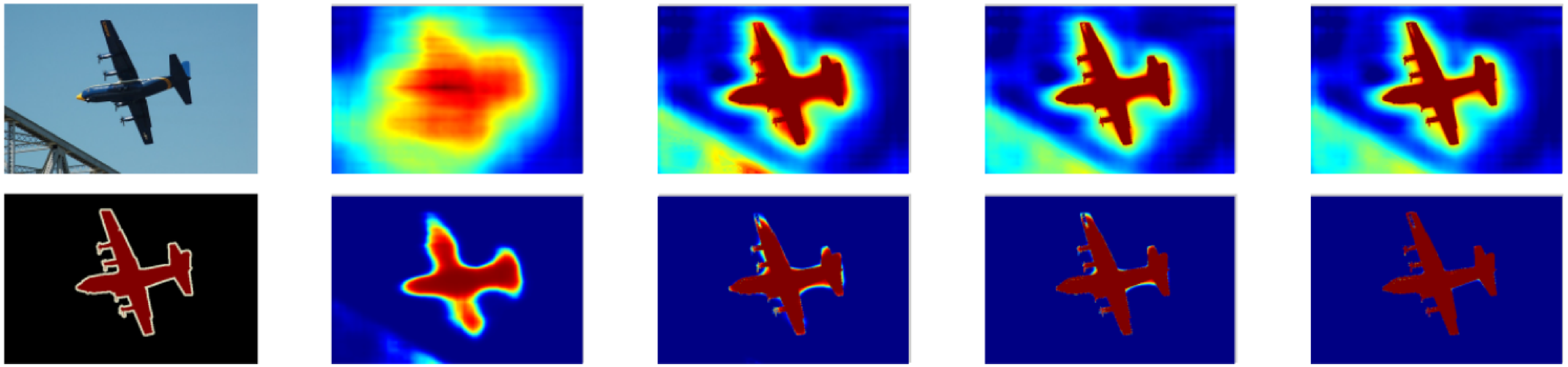



Figure 12: Score map (input before softmax function) and belief map (output of softmax function) for aeroplane. We show the score (1st row) and belief (2nd row) maps after each mean field iteration. The output of the last CNN layer is used as input to the mean field inference.